📄 Mathematical Paper: "The Theory of Forced Networks for Probabilistic Verification"

The Theory of Forced Networks for Probabilistic Verification

Abstract: We present an abstract mathematical framework for an information system that guarantees tamper detection through a small random sample inspection. This is achieved by constructing a forced network of dependencies between entities, causing errors to propagate in a controlled and calculable manner.

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1. Core Definitions

Definition 1.1: Information System

We define an **information system** as a tuple:

$$\mathcal{S} = (V, E, \mathcal{H}, \phi)$$

where:

- $V = \{v_1, v_2, \ldots, v_n\}$: the set of entities (e.g., transactions, records, etc.)
- $E \subseteq V \times V$: the set of linkage relations between entities
- $\mathcal{H}$: the hash space
- $\phi : V \to \mathcal{H}$: the hash function that maps each entity to its hash

Definition 1.2: Pollution Hash

For an entity v_i, we define the **pollution hash** H_i as:

$$H_i = \mathrm{Hash}\left(\mathrm{data}(v_i) \parallel \backslash\mathrm{bigparallel}_{j \in N(i)} \phi(v_j)\right)$$

where:

- $N(i) = \{j \mid (i,j) \in E\}$: the neighbors of v_i
- $\|$: concatenation operator
- $\backslash\mathrm{bigparallel}$: concatenation of all neighbor hashes, sorted (to ensure determinism)

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2. Core Idea: Principle of Forced Error Propagation

Theorem 2.1: Pollution Propagation

Premise: Suppose entity v_t has been tampered with, changing $\phi(v_t)$ to $\phi'(v_t) \neq \phi(v_t)$.

Conclusion: For every entity v_i such that $N(i) \cap \{t\} \neq \emptyset$ (i.e., v_t is a neighbor), we have:

$$H_i' \neq H_i$$

where H_i' is the recomputed hash after tampering.

Proof:

Assumptions:

1. Hash is a collision-resistant cryptographic function.
2. H_i directly depends on $\phi(v_t)$ if $t \in N(i)$.

Proof by contradiction:
Assume $H_i' = H_i$ despite the change in v_t.
This implies:

$$\mathrm{Hash}(\cdots \| \phi(v_t) \| \dots) = \mathrm{Hash}(\cdots \| \phi'(v_t) \| \dots)$$

This contradicts the collision-resistance property of Hash.
Therefore, $H_i' \neq H_i$. ∎

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3. Probabilistic Detection Theory

Definition 3.1: Impact Set

For a tampered entity v_t, define:

$$\text{Impact}(t) = \{t\} \cup \bigcup_{i:t \in N(i)} \{i\}$$

This is the set of entities whose hashes become invalid after tampering with t.

Definition 3.2: Detection Probability

When inspecting a random sample $S \subseteq V$ of size $|S| = m$:

$$P_{\text{detect}} = \mathbb{P}[S \cap \text{Impact}(t) \neq \emptyset]$$

Theorem 3.1: Lower Bound on Detection Probability

Premise:

- Each entity has exactly k neighbors ($|N(i)| = k$ for all i)
- Random sampling of size m
- $p = \frac{|\text{Impact}(t)|}{|V|} = \frac{k+1}{n}$

Conclusion:

$$P_{\text{detect}} \geq 1 - \left(1 - \frac{k+1}{n}\right)^m$$

Proof:

The probability that the sample misses all elements of Impact(t) is:

$$\mathbb{P}[\text{miss}] = \left(1 - \frac{k+1}{n}\right)^m$$

Since success and failure are complementary:

$$P_{\text{detect}} = 1 - \mathbb{P}[\text{miss}] = 1 - \left(1 - \frac{k+1}{n}\right)^m$$

(For large n, this approximates $1 - e^{-m(k+1)/n}$.) ∎

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4. Optimal Case: Exponential Propagation

Definition 4.1: Multi-Layer System

To achieve wider propagation, construct a layered system with L layers:

$$\mathcal{S}_L = (\mathcal{S}_{L-1}, E_L, \phi_L)$$

where each layer connects to the previous one via a function $f : \mathcal{H}_{L-1} \to E_L$.

Theorem 4.1: Detection Lower Bound in Multi-Layer Systems

In an L-layer system where each entity in layer ℓ connects to k_ℓ entities in layer $\ell - 1$:

$$P_{\text{detect}} \geq 1 - \prod_{\ell=1}^{L} \left(1 - \frac{k_\ell + 1}{n_\ell}\right)^{m_\ell}$$

where m_ℓ is the sample size taken from layer ℓ.

Proof: By induction over independent layers and applying Theorem 3.1 to each. ∎

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5. Application: Financial Transaction System

5.1 Abstraction

For a transaction $T_i = (\text{from}_i, \text{to}_i, \text{amt}_i, \tau_i)$:

- $V = \{T_1, \dots, T_n\}$
- $N(i)$: transactions sharing the same account or close timestamps
- $\phi(T_i) = \mathrm{sha256}(\mathrm{from}_i \parallel \mathrm{to}_i \parallel \mathrm{amt}_i \parallel \tau_i)$

5.2 Practical Proof

Given:

- $n = 10^6$ transactions
- $k = 5$ links per transaction
- $m = 1000$ random transactions (0.1% of the system)

Single-layer calculation:

$$P_{\mathrm{detect}} = 1 - \left(1 - \frac{6}{10^6}\right)^{1000} \approx 1 - e^{-0.006} \approx 0.006 \quad (0.6\%)$$

Improvement with $L = 3$ layers and increasing connectivity (e.g., hierarchical buckets):

- Layer 1: $n_1 = 10^6, k_1 = 5, p_1 = 6 \times 10^{-6}, m_1 = 1000$
- Layer 2: $n_2 = 10^4, k_2 = 50, p_2 \approx 5.1 \times 10^{-3}, m_2 = 50$
- Layer 3: $n_3 = 10^2, k_3 = 500, p_3 \approx 5.01, m_3 = 5$

Then:

$$P_{\mathrm{detect}} \geq 1 - (0.999994)^{1000} \cdot (0.9949)^{50} \cdot (0.5)^{5} \approx 0.97 \quad (97\%)$$

Result: With inspection of ~0.1% of the base layer and negligible fractions of upper layers, detection reaches ~97%.

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6. Mathematical Summary

Master Theorem

Premise: In any information system where:

1. Entities are linked with degree $k \geq 3$
2. Hashes depend on data and neighbors
3. Inspection is via unbiased random sampling

Conclusion: To detect a single tamper with probability $\epsilon > 0.95$, it suffices to inspect:

$$m \geq \frac{\ln(1/(1-\epsilon))}{\ln\left(1 - \frac{k+1}{n}\right)} \approx \frac{c \cdot n}{k+1}$$

For large n, this is $O(n/k)$. To reach $m = O(\log n)$, k must be $\Omega(n/\log n)$.

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7. Limitations and Improvements

7.1 Minimum Connectivity Requirement

$k = \Omega(\log n)$ is insufficient to avoid graph hashing issues; $k = \Omega(n/\log n)$ is required for logarithmic sampling.

7.2 Trade-off Between Accuracy and Performance

$$\text{Objective} = \min_{k,m} \left(\alpha \cdot \text{Cost}(m) + \beta \cdot (1 - P_{\text{detect}})\right)$$

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8. Conclusion

We have provided a rigorous mathematical proof that, through:

1. **Forced network construction** between entities
2. **Chained dependency hashing**
3. **Probabilistic sampling**

one can achieve **efficient fraud detection** by inspecting only a small subset of data, reducing complexity from $O(n)$ to sublinear (potentially $O(\log n)$ with dense connectivity), while maintaining strong security guarantees.

Notation Glossary:

- $O(\cdot)$: Big-O notation (asymptotic upper bound)
- $\Omega(\cdot)$: Big-Omega notation (asymptotic lower bound)
- $||$: Concatenation
- $\mathrm{Hash}(\cdot)$: Collision-resistant hash function
- $\mathbb{P}[\cdot]$: Probability